

Page 1 - Introduction

OmniBrain is an AI-powered document intelligence platform designed to help users search, understand, and interact with large collections of documents using Retrieval-Augmented Generation (RAG). The platform combines document parsing, semantic search, vector embeddings, and large language models to generate accurate answers. OmniBrain was founded for multimodal document intelligence. Its primary goal is to allow users to upload PDFs and ask natural language questions about their contents. Instead of relying only on a language model's knowledge, OmniBrain retrieves relevant document chunks from a vector database before generating a response. The platform supports scalable document indexing, efficient retrieval, and contextual answer generation while maintaining references to the original document pages.

Page 2 - System Architecture

The architecture of OmniBrain consists of several components working together. Uploaded PDF documents are parsed into pages and split into smaller text chunks. Each chunk is converted into embeddings using the nomic-embed-text embedding model provided by Ollama. These embeddings are stored in a persistent ChromaDB vector database. When a user asks a question, the system embeds the query, performs semantic similarity search, retrieves the most relevant chunks, and constructs a context for the language model. The language model used by OmniBrain is llama3.2 through LangChain's ChatOllama integration. The generated answer is based only on the retrieved context, improving accuracy and reducing hallucinations.

Page 3 - Testing Information

This document is intended for testing the complete RAG pipeline. Expected verification questions include: 1. What was OmniBrain founded for? 2. Which embedding model is used? 3. Which vector database stores the document embeddings? 4. Which language model generates the answers?

Expected answers: • OmniBrain was founded for multimodal document intelligence. • The embedding model is nomic-embed-text. • ChromaDB stores the document embeddings. • The language model is llama3.2. If the RAG pipeline is functioning correctly, the answer should be generated from the retrieved document context and include the correct page number in the sources.